CHAPTER

24

Color Perception

Restorative dentistry is a blend of science and art. The success of restorative dentistry is determined on the basis of functional and esthetic results. To achieve esthetics, four basic determinants are required in sequence; viz. position, contour, texture and color. Color makes an object more appealing, attractive and gives the pleasure of observing. *Color is a psychophysical sensation provoked by eye in the presence of light and interpreted by the brain*. Color combination not only improves esthetics, but also makes the restoration appear natural and attractive. The basic fundamentals of color and light, the radiation spectrum and the optical characteristics of the object are to be understood before evaluating and selecting proper color shade for the restoration.

SOURCE OF LIGHT

Light is mandatory for any color to exist. Light is a form of electromagnetic energy, which is a small part of the large and continuous radiation spectrum (Fig. 24.1). Sunlight or the light of a glowing body is not single type of wavelength but a combination of different wavelengths (measured in nanometers, i.e. millionths of a millimeter). Light in the wavelength range of 380 to 760 nm is referred to as the visible light. Wavelengths less than 380 nm are in the ultraviolet range (cosmic rays, gamma rays and X-rays). Wavelengths above 760 nm are in the infrared range (radar waves, TV waves and radio waves).

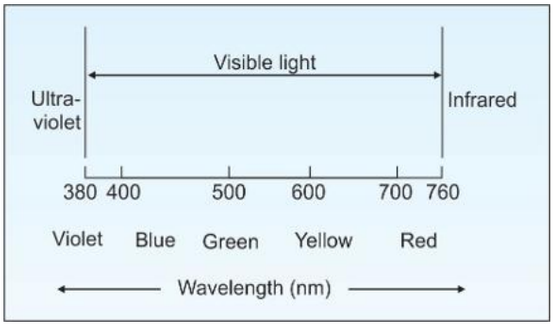


Fig. 24.1: Radiation spectrum

Table 24.1: Wavelength (nm) range of different colors

Color	*Wavelength range (nm)*
Violet	380–450
Blue	450–480
Blue-green	480–510
Green	510–550
Yellow-green	550–570
Yellow	570–590
Orange	590–630
Red	630–750

1 nm = 0.000001 mm

A beam of light when passed through a glass prism gets converted into its component parts depending upon the wavelength. They are viewed as a series of glowing spectral colors ranging from violet, blue, green, yellow and red without any clear demarcation between them. White light is produced when all these colors are combined together. The wavelength range of different colors is shown in Table 24.1.

OPTICAL CHARACTERISTICS

When light falls on an object, it is either reflected, transmitted or absorbed. If the whole light is reflected, the object appears white; whereas, if the light is entirely absorbed, the object appears black. Greater the intensity of the light source, or greater the amount of light reflected, higher is the brightness of the object being viewed. The final effect is determined by the transparent, translucent or opaque characteristics of the object.

The optical features are as follows:

a. Reflection

Reflected rays are those rays, which bounce back from the surface of the object being hit, instead of being transmitted or absorbed. Reflection can be of two types:

i. *Specular reflection:* The light is reflected from the point of incidence in a single direction (Fig. 24.2).

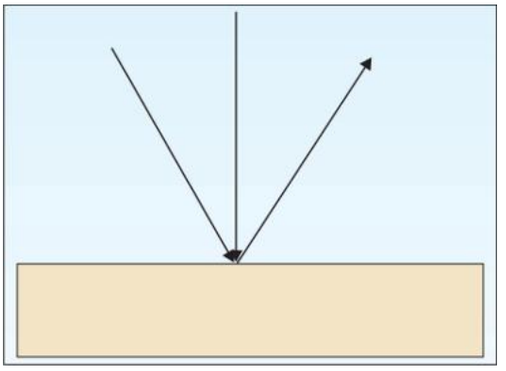


Fig. 24.2: Specular reflection

Such type of reflection is seen on perfectly smooth surfaces (mirror reflection). These reflections do not provide any information about color.

ii. *Diffuse reflection:* The light is reflected from the point of incidence in various directions at angles different from the angle of incidence (Fig. 24.3). Diffuse reflection is seen on rough surfaces. These reflections carry information about the color of an object.

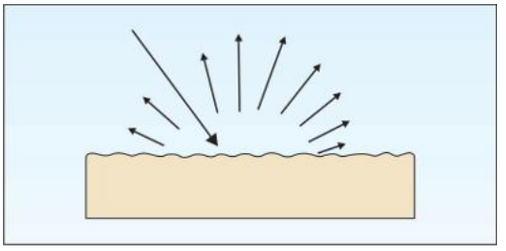


Fig. 24.3: Diffuse reflection

b. Transmission

A number of physical and chemical processes account for 'emission' of light from a source. No light source can emit perfectly white light (i.e. exactly the same amount of wavelength).

Transmission occurs when light passes through a transparent/translucent material. If the material is completely transparent, all light is transmitted and the color perceived in 'white'. In case the material is completely opaque, all light is absorbed and the color perceived is 'black'. However, in most cases, some wavelength (color) is absorbed and the other is transmitted. When light emerges through an object, the process is called 'transmission' (Fig. 24.4).

c. Refraction

When light travels from one medium into another medium, the velocity changes and the light ray is bent. This process is known as 'refraction' and the ratio

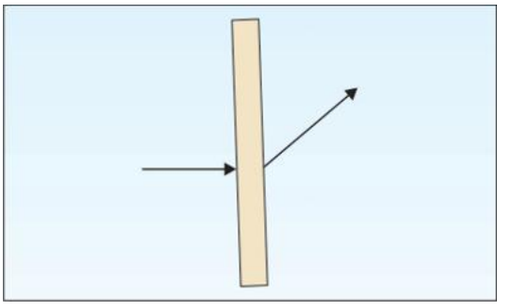


Fig. 24.4: Transmission

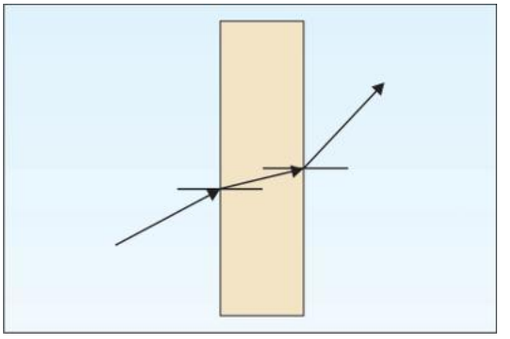


Fig. 24.5: Refraction

between velocities is known as 'index of refraction' (Fig. 24.5).

d. Transparency

A medium is 'transparent' when it allows maximum transmission of light without any distortion (Fig. 24.6). A clear window glass through which the object can be viewed distinctly is considered as transparent.

e. Translucency

A medium is 'translucent' when it allows transmission of light in a diffused manner (Fig. 24.7). A frosted glass through which an object is viewed indistinctly is considered as translucent. Diffuse transmission of light is because of the enormous number of internal

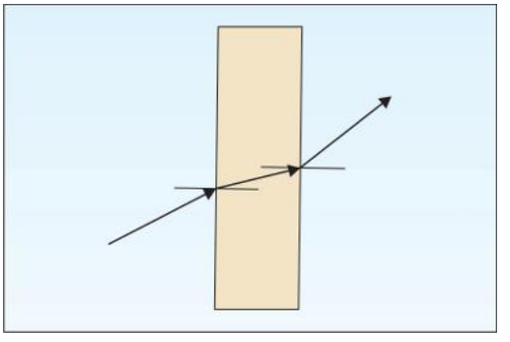


Fig. 24.6: Transparent

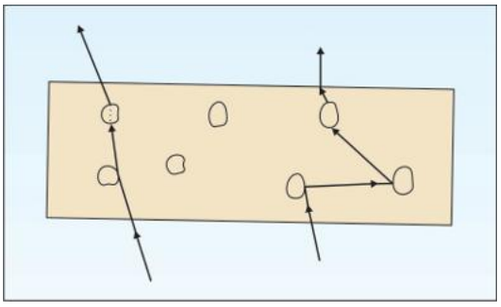


Fig. 24.7: Translucent

reflections, external reflections and the refractions as well.

f. Opacity

A medium is said to be opaque when it transmits no light (Fig. 24.8); for example, a brick wall. The incident light is either refracted or absorbed depending upon the nature of the medium. The degree of translucency can be defined more accurately by measuring the degree of opacity. A body that transmits no light is said to be 100% opaque; whereas, one which transmits, (say 30%) light is said to be 70% opaque. In dentistry, the translucent and opaque characteristics are important than the transparency.

g. Light Scattering

Whenever light is transmitted through a body, the intensity of the beam is reduced, because of the fact that part of the energy is refracted at each point of change in medium (Fig. 24.9). For example, consider light striking a window glass. Part of it is reflected on the outside surface and the rest is refracted towards the glass pane. As the light leaves the pane to enter air, another portion is reflected and the remainder is refracted in the air. When the medium is heterogeneous, the number of reflections and refractions occurring inside the medium is considerably increased. This phenomenon is known as 'scattering of light'. The amount of light scattered depends upon the difference in refractive indices of the dispersed and the dispersion medium. Scattering within limits produces translucency in a transparent

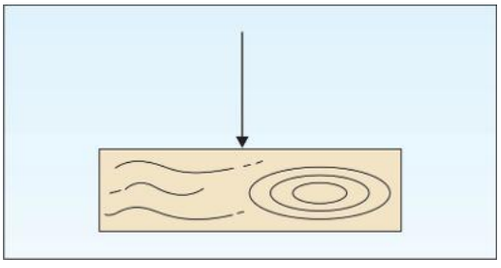


Fig. 24.8: Opaque

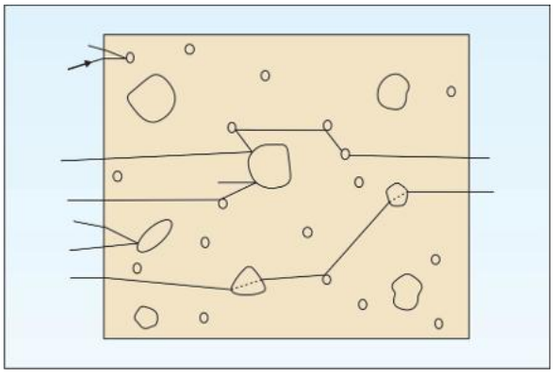


Fig. 24.9: Scattering of light due to several internal reflections and refractions

medium, while an excessive scattering produces opacity.

h. Surface Gloss

The amount of specular and diffuse reflection occurring on a surface is determined by its smoothness. Ceramic surfaces have slight irregularities; hence, the reflection is a combination of specular and diffuse types. On a glazed surface, specular reflection predominates; whereas, on a rough surface, diffuse reflection predominates (Fig. 24.10A, B).

COLOR

Color is the property of a surface or substance due to the absorption of certain light rays and reflection of others within the range of wavelengths (380 to 760 nm) adequate to excite the retinal receptors. Briefly, the color is a result of selective absorption and selective reflection. For example, an object appears red when it absorbs all constituent lights of the spectrum except red, which is reflected back and reaches the eye. Color perception is psychophysical process—'a psychological response to a physical stimulus, i.e. it involves both objective and subjective phenomena'. Objective is the falling of light rays on the retina and subjective is its interpretation by the

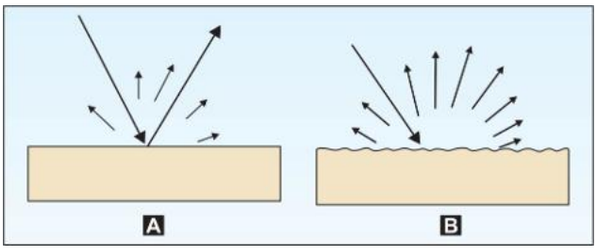


Fig. 24.10: Surface gloss: **(A)** Predominance of specular reflection on smooth glazed surface; **(B)** Predominance of diffuse reflection on rough surface

brain. Since color determination involves a psychological response, it is evident that an object may appear to be differently colored when viewed by different observers.

STRUCTURE AND FUNCTION OF EYE

The main features of human eye are shown in Fig. 24.11. Light enters eye through the cornea. Immediately behind cornea is the iris. It is a fibrous membrane with central circular aperture, which contracts or expands to control the amount of light entering the eye. This compensating property of the eye is known as chromic adaptation. Eye lens is situated behind the iris and has sufficient refracting power to focus the image of the object on the retina. Ciliary muscles control the focal length of the eye lens. Various layers of the retina are shown in Fig. 24.12. From inside to outside, these consist of pigment cells, photosensitive receptors, i.e. rods (rod shaped) and cones (cone shaped), bipolar cells, ganglion cells and optic nerve.

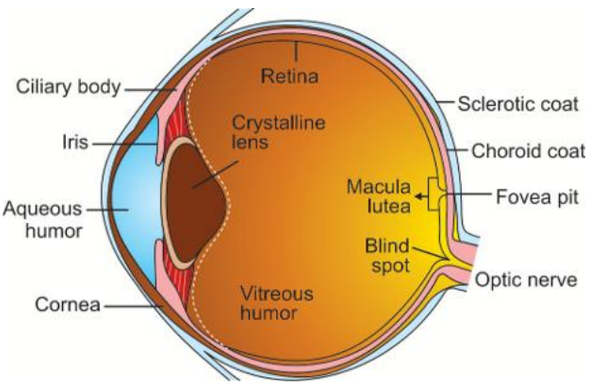


Fig. 24.11: The human eye

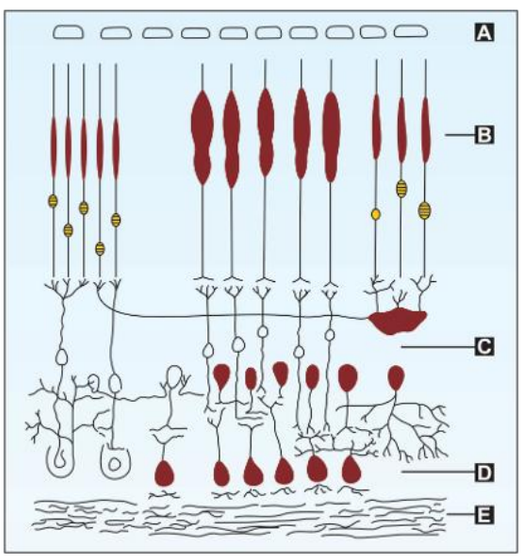


Fig. 24.12: Structure of retina: **(A)** Pigment layer; **(B)** Rods and cones; **(C)** Bipolar cells; **(D)** Ganglion cells; **(E)** Optic nerve

PERCEPTION OF COLOR

Irrespective of the nature of light source, color results only from a reflected or transmitted beam of white light or a portion of it. There should be enough light to elicit a color. No object appears colored at low levels of illumination. This is because at low lights, the more sensitive rods get functional, which are unable to differentiate color.

Color perception involves series of events, viz. (a) light source, (b) surface viewed, (c) individual observer (eye and brain), and (d) surrounding environment under which the object is being viewed (Fig. 24.13). All these features are interrelated and a variation in any may affect the color perception.

a. Light Source

Light source should be without the dominance of any wavelength. [If an object is viewed under particular wavelength (color bands), the specific color becomes dominant to the observer]. Different types of lights and their designation by CIE (Commission Internationale de l'Eclairage) are summarized in Table 24.2.

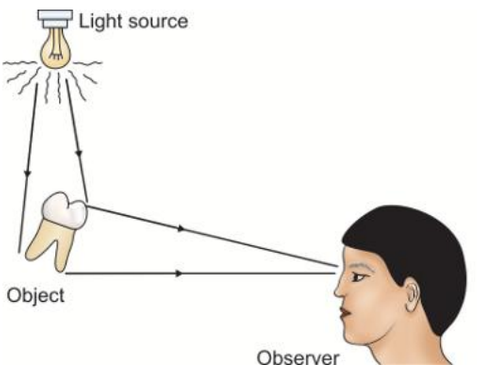


Fig. 24.13: Perception of color

Table 24.2: Different types of lights and their designation as proposed by CIE (Commission Internationale de l'Eclairage)

Light type	*Color temperature K*	*Light source*
A	2856	Incandescent bulb
B	4860	Midday sunlight
C	6770	Average northern sunlight
D	6500	Average daylight (overcast)
E	5455	Equal energy radiator
F	3000	Fluorescent tube 'warm-white'

Sunlight is the most preferred light for shade selection. However, it may vary in its spectral distribution throughout the day depending upon the time and weather. Average daylight (intensity 1500 Lx) is considered ideal for shade determination. Light at dawn and dusk is rich in the red-yellow spectrum. Fluorescent lamp (gas discharge lamp) includes narrow band widths of wavelengths that are elicited by the atoms of the respective gas within the tube. Incandescent (tungsten) bulbs show a spectrum of wavelengths that are dominant in the orange red energy.

b. Surface Viewed

The surface of an object being viewed affects color determination via absorption, reflection and transmission. The surface is analyzed with a spectrophotometer. (The percentage of reflectance compared with the total amount of light incident on the surface is recorded.)

Three types of light sources are:

i. *Incandescent light:* Emits high concentration of yellow waves (not suitable for shade matching; low color rendering index).
ii. *Fluorescent light:* Emits high concentration of blue waves (not suitable for shade matching; average color rendering index).
iii. *Daylight:* Northern daylight closely emits full spectrum of white light (used in color matching; best color rendering index).

c. Individual Observer (Eye and Brain)

Eye is most sensitive to brightness in wavelength between 540 and 570 nm (yellow-green); thereon, the brightness diminishes in both directions. Interpretation of color by the brain differs for different observers. The color perception is also affected by the eye fatigue. Prolonged viewing of a color for 15–30 seconds strains the cones and causes depletion of photopigments. The cones can be made sensitive to the desired color by gazing at the complementary color.

d. Surrounding Environment

The surrounding environment also influence the interpretation of the color, viz. background, wet/dry conditions, angle and intensity of illumination, etc.

ADDITIONAL FEATURES IN COLOR PERCEPTION

The additional features in color perception are:

a. Metamerism

Metamerism is defined as "*the phenomenon when color of two objects appears to match under one set of viewing conditions but differ under different sets of conditions, like a change in light source or the observer*". Metamerism should not be confused with color changes that occur when a single object is viewed under different types of lights or by different observers. In metamerism, there should always be present two objects whose spectrophotometric curves are different. An exact match is almost impossible when two different materials are used (e.g. natural tooth enamel and porcelain) but the more closer the curves are matched, the fewer are the metameric problems.

Various forms of metamerism are:

- Illuminant metamerism (due to different light conditions)
- Observer metamerism (due to difference in interpretation by different observers)
- Geometric metamerism (due to change in angle of view)
- Field size metamerism (due to change in area of view)

b. Fluorescence

Fluorescence is the absorption of radiation of a particular wavelength and its re-emission as radiation of longer wavelength.

Natural tooth has fluorescing qualities because of its organic content [absorbs wavelengths between 300 and 380 nm (ultraviolet range)]. Sunlight, photoflash lamps, UV lights, etc. are sources of ultraviolet radiations. The energy absorbed by the tooth is then released as radiations of longer wavelengths, generally in the range of 400–500 nm (yellow white fluorescence). Because of fluorescence, the human tooth has the brightness and life-like appearance. Any restoration that is to replace natural tooth structure should contain fluorescing agents. It is important to match the shade under two or more lighting conditions, out of which one should have adequate UV component (incandescent lights).

c. Opalescence

Opalescence is the phenomenon in which a material appears to be of one color when light is reflected from it and of another color when light is transmitted through it. Opals act like prisms and refract different wavelengths to varying degrees. The shorter wavelengths refract more to escape an optically dense material than the longer wavelength. The hydroxyapatite crystals of enamel also act as prisms. Wavelengths of light have different degrees of translucency. When illuminated, opals and enamel will transilluminate the reds and scatter the blues within its body; thus, enamel appears bluish even though it

is colorless. The opalescent effects of enamel brighten the tooth and give it optical depth and vitality.

BASIC COLOR SCHEMES

The color wheel or color circle (Fig. 24.14) is the basic tool for combining colors. The first color circle was designed by Sir Isaac Newton in 1666. The accepted circle is a wheel of 12 colors; three primary colors (red, yellow and blue), three secondary colors (green, orange and violet), created by mixing two primary colors and six tertiary colors created by mixing primary and secondary colors (Table 24.3).

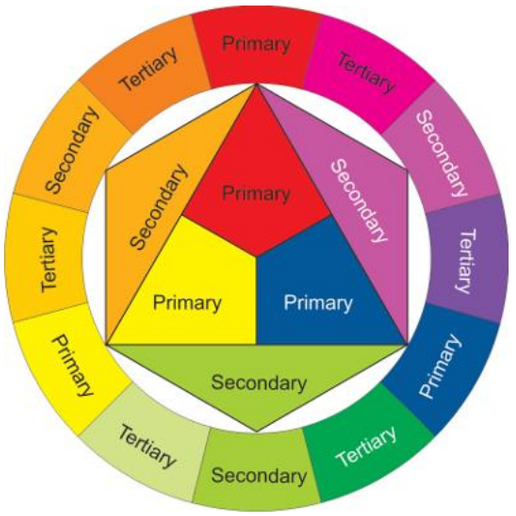


Fig. 24.14: The color wheel

Table 24.3: Basic color schemes

Primary color	*Complementary color*
• Red	• Green
• Yellow	• Violet
• Blue	• Orange

COLOR HARMONIES/COLOR CHORDS

The color combinations that are considered pleasing are called color harmonies or color chords. They consist of two or more colors having fixed relation in the color wheel.

These are:

- Complementary color scheme (Colors that are opposite to each other on the color wheel are considered to be complementary colors; for example, red and green).
- Analogous color scheme (Colors that are next to each other on the color wheel).
- Triadic color scheme (Three colors that are evenly spaced around the color wheel).
- Tetradic or rectangular color scheme (Four colors arranged into two complementary pairs).
- Split complementary color scheme (In addition to the base color, it uses two colors adjacent to its complement).
- Square color scheme (Similar to rectangle, but all four colors spaced evenly around the color circle).

The combining of colors is explained by following theories.

a. Additive Color Theory

The additive primary colors are red, green and blue (RGB). Combining one additive primary colors with equal amounts of other results in the additive secondary colors of cyan, magenta and yellow (Fig. 24.15). Combining all three additive primary colors in equal amounts will produce the color white. Remember combining additive colors creates lighter colors; so adding all three primary colors results in a color so 'light', [seen as white (Table 24.4)]. By changing ratio of the three primary colors by varying degrees, a wide range of colors can be produced (Table 24.5).

b. Subtractive Color Theory

The subtractive primary colors are cyan, magenta and yellow (Fig. 24.15). Subtractive color mixing occurs when light is reflected off a surface or is filtered through a translucent object. For example, a red surface only appears red, because it absorbs (subtracts) all the light that is not red and reflects only the red light.

Table 24.4: Additive colors combined in equal parts

Blue + Green	=	Cyan
Red + Blue	=	Magenta
Green + Red	=	Yellow
Red + Green + Blue	=	White

Table 24.5: Additive colors combined in unequal parts

1 Green + 2 Red	=	Orange
1 Red + 2 Green	=	Lime
1 Green + 1 Blue + 4 Red	=	Brown

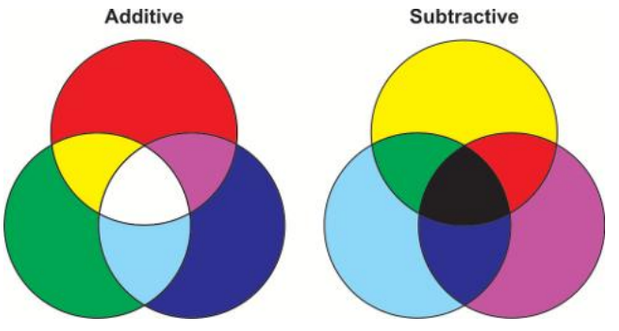


Fig. 24.15: Additive and subtractive color mixing

Mixing of two subtractive primary colors results in a color, which is complementary to the remaining primary. For example, if cyan and magenta are mixed, it will form blue which is complementary to the yellow (third subtractive primary color) (Table 24.6).

Table 24.6: Subtractive colors mixing

Combine	*Absorbs*	*Leaves*
Cyan + Magenta	Red + Green	Blue
Cyan + Yellow	Red + Blue	Green
Magenta + Yellow	Green + Blue	Red
Cyan + Magenta + Yellow	Red + Green + Blue	Black

COLOR SYSTEMS

Color, perceived when light reflected from an object enters the eye, consists of 'quantity' and 'quality'. 'Quantity' is the variable lightness (value) and is the total amount of light reflected from an object (indicated by the relative nearness of a color to white or black on an achromatic gray scale, such as black and white photograph). 'Quality' is the chromatic component of color consisting of two variables, which can be represented in two dimensions.

The systems describing color of tooth are as follows.

1. Munsell Color System

Munsell color system defines color by hue, saturation and lightness. The most widely used system arranges the three dimensions of the color in equal visual steps.

The Munsell color axis simulates a cylinder (Fig. 24.16). The Munsell cylinder is formed by a series of wheels stacked one upon the other. The hub of each wheel represents the 'value axis'. A colorless or achromatic axis extends through the center of the cylinder; pure white at the top (assigned a value of 10) and pure black at the bottom (assigned a value of 0). A series of greys progressing from black to white in equal steps connects these two extremities. Hues are arranged sequentially around the rim of the wheel. There are 100 hues in this scale [10 major hues (5 principal and 5 intermediate hues)]; red (R), yellow (Y), green (G), blue (B) and purple (P) are the principal hues; whereas, yellow-red (YR), green-yellow (GY), blue-green (BG), purple-blue (PB) and red-purple (RP) are the intermediate hues. Within each hue, the colors are arranged in scales according to their lightness/darkness (value) and their purity (saturation). Colors with lower value numbers are known as 'dark colors' and are arranged towards the bottom. Conversely, colors with higher value numbers are known as 'light colors' and are arranged towards the top.

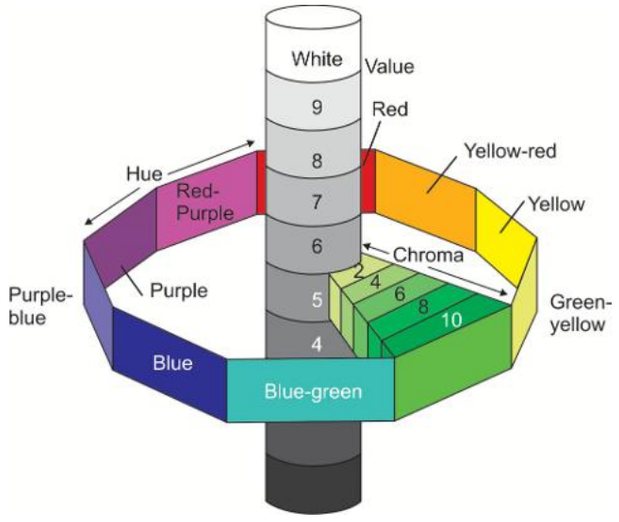


Fig. 24.16: Munsell color system

The spokes of the wheel represent the saturation (chroma) of a color. The chroma scale starts from zero and extends to a maximum of fourteen.

The designation of any particular color located in the Munsell color axis is given by the notation HV/C, where H stands for hue, V for value and C for chroma, e.g. 8R3/7 would mean hue is 8 (red), value is 3 and chroma is 7. The scale of value extends from 0/, 1/, 2/ …. 10/ and chroma from /0, /1, /2 …../14 or farther. Most of the dental shade guides are based on a section of the Munsell color system that includes color range of the natural teeth.

Munsell color is defined in three dimensions—hue, chroma and value. (*Value is determined first, followed by chroma; hue is described last by matching with shade tabs of value and chroma*).

Hue

Hue distinguishes one color family from another, e.g. red, yellow, blue, etc. (type of color) (Fig. 24.17A). It is specified as the dominant range of wavelengths in the visible spectrum that yield the perceived color. Hue is represented by A, B, C and D on the commercial vita shade guide.

Chroma

Chroma distinguishes a strong color from a weak one [saturation/intensity of a hue (Fig. 29.17B)]. This dimension can be more easily explained by employing the concept of painting a box. Suppose one side of the box is painted in pure red. Now before painting the second side, a little amount of gray is added to the previous red; the color perceived is weak red. Similarly, while painting the third side, more amount of gray is added; the color becomes more weak. Adding gray reduces the chroma, while hue remains the same.

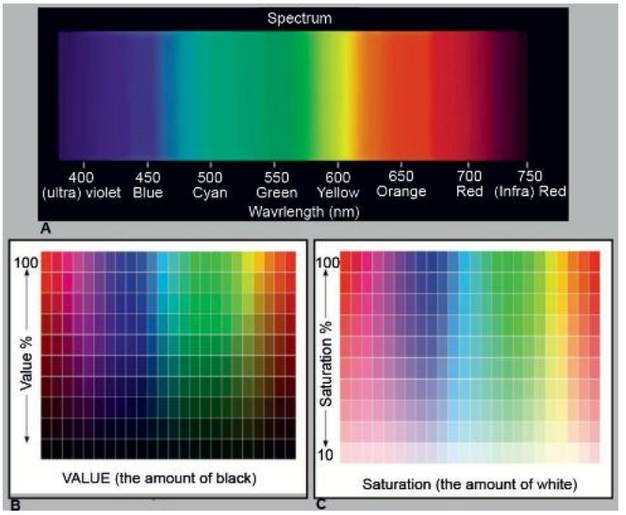


Fig. 24.17: Dimensions of color: **(A)** Hue; **(B)** Value; **(C)** Chroma

Value

Value, the amount of light returned from an object, distinguishes whiteness from blackness. A color of high value is transmitted as light gray and that of a low value as dark gray (Fig. 24.17C). If a series of colors have the same values, all of them would be composed of the same grays. A younger tooth is generally whiter and brighter than its older counterpart. Increase in the thickness of opaque or core porcelain raises the value; whereas, an increase in enamel (translucent) porcelain reduces the value. Lowering value means less light returns from the illuminated object and the rest is being absorbed or scattered elsewhere.

The value of a specific color is the value of gray added to the original color. If the gray of same value is added each time, no change in value occurs, only the chroma reduces. On adding a value higher than the original value of the color, the resulting color has the same hue, reduced chroma and higher value. On adding a value lower than the original value, the resulting color has the same hue, reduced chroma and reduced value.

2. CIE System

The Commission Internationale de l'Eclairage (CIE) described color concept based on the theory that eye possesses receptor for three primary colors (red, green and blue) and all other colors are seen as mixture of these three primary colors.

CIE introduced instrumental color measurement based on the rationale that a color sensation can be matched by the mixing of three primary wavelengths (red, green and blue; represented by [R], [G] and [B]).

In CIE system, a series of (human) observers were employed, using a 2°viewing angle. The observers determined the amount of the three primaries that were needed to match all spectral colors within the visible spectrum. The results obtained for the observers were collected and the concept of a standard observer was introduced.

Primaries: The three imaginary primaries are specified as [X], [Y] and [Z].

Color matching functions: These characterize the response of the observer, denoted as $\bar{x}$, $\bar{y}$ and $\bar{z}$.

Standard observer: Earlier 2° viewing angle was used; however, a 10° angle is accepted in principle.

Tristimulus values: The amount of the three primaries (i.e. R, G and B) are known as the tristimulus values of the color [C]. These are denoted as X, Y and Z. The color of an object depends on:

- Spectral energy distribution of the light that illuminates the sample (S)
- Reflectance characteristics of the sample (R)
- Sensitivity of the observer (color matching functions).

The tristimulus value is the amount of a primary that will match the color of a given object. The given color is matched by:

$$X[X] + Y[Y] + Z[Z]$$

As color of that object also depends on the characteristics of the illuminant, object and observer, it follows that the tristimulus values are related to S, R and $\bar{x}$, $\bar{y}$ and $\bar{z}$. For a given wavelength:

$$X = S_\lambda R\bar{x}_\lambda$$
$$Y = S_\lambda R\bar{y}_\lambda$$
$$Z = S_\lambda R\bar{z}_\lambda$$

3. CIE (Modified) L*a*b* Color Space (CIELAB)

The modified versions (CIELAB) are mostly used in dentistry [color difference is marked as ΔE, which denotes: Symbol delta (Δ) signifies difference and E is the first letter of German word meaning 'sensation'].

CIE L*a*b* (three-dimensional coordinates)	
L*: Lightness (value): Relative darkness or lightness. The greater the total amount of light reflected, higher is the value.	
a*: Green-red coordinate:	–a* = green +a* = red
b* : Blue-yellow coordinate :	–b* = blue +b* = yellow

The tristimulus values of the sample are transformed into L*, a* and b* coordinates as mutually perpendicular axes. In this three-dimensional color space, lightness is represented by L* on a scale of 0

for black to 100 for white. The hue and chroma of a sample are represented on an a* versus b* plot. a* is the red/green coordinate with +a* indicating red and –a* indicating green. b* is the yellow/blue coordinate, with +b* indicating yellow and –b* indicating blue (Fig. 24.18).

The CIELAB color difference is expressed as:

$$\Delta E^{*ab} = \sqrt{(L^*_2 + L^*_1)^2 + (a^*_2 + a^*_1)^2 + (b_2 + b_1)^2}$$

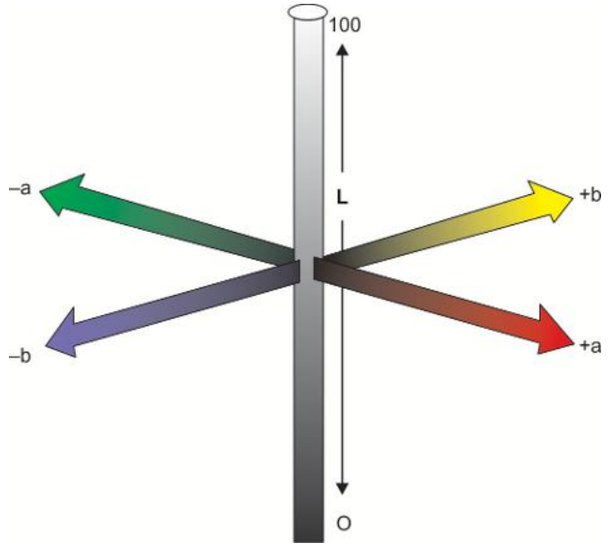


Fig. 24.18: CIE L*a*b* color space

4. CIE (Modified) L*C*h° Color Difference (CIELCH)

The coordinates, L*, a* and b* are transformed into L*, C* and h°. The lightness (L*) coordinate is the same as in L*, a*, b* space, with the chroma (C*) coordinate being perpendicular to the lightness axis. The hue angle (h°) is expressed in degrees (0° at the +a* axis, 90° at the +b* axis, 180° at the –a* axis, 270° for –b* and back to 360° = 0°).

The CIELCH color difference is expressed as:

$$\Delta E^{*ab} = [(\Delta L^*)^2 + (\Delta C^*)^2 + (\Delta H^*)^2]^{0.5}$$

OPTICS OF NATURAL TEETH

Color of natural tooth is a result of light reflected directly from the tooth surface combined with light reflected from the dentin and a minor contribution from the oral background. Dentin is the primary source of color and the rays reflected from it are transmitted via enamel (modified by the thickness and degree of enamel translucency). The effect of enamel thickness and translucency on tooth color is evident in the incisal, middle and cervical thirds of the tooth,

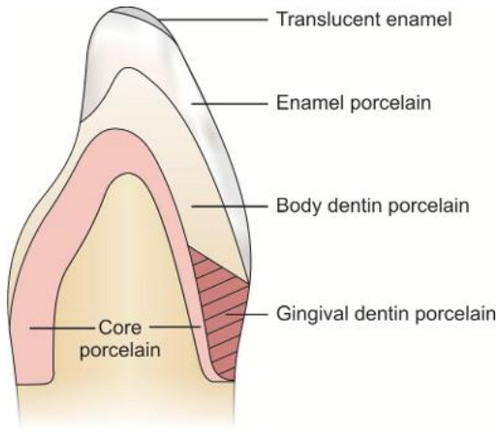


Fig. 24.19A: The gingival, dentin and enamel colors along with the tracing of enamel outline



Fig. 24.19B: Translucent, opaque, enamel and dentin porcelain powders

which needs to be duplicated in porcelain restorations (Fig. 24.19A) using translucent, opaque, enamel and dentin porcelain (Fig. 24.19B).

a. ***Incisal third of tooth:*** This is primarily composed of translucent enamel; the shade may vary from light to dark-gray to bluish. The translucency and the grayness may extend into approximal areas and sometimes into dentin. The color of incisal third is referred to as the 'enamel color'.
b. ***Middle third of tooth:*** This area is primarily composed of dentin. The color of the middle third region is determined by the dentin hue having been modified by the overlying translucent blue gray enamel. It generally falls in the range of orange-red, blue-yellow or gray-yellow. The color of middle third is referred to as the 'body dentin color'.
c. ***Cervical third of tooth:*** In this area, the enamel is thin and the underlying dentin color is quite marked. The color of the cervical half region is referred to as the 'gingival dentin color' and is in the range of orange-yellow to brown.

DENTAL SHADE GUIDES

Dental shade guides are a collection of various color combinations available from the manufacturers; which aid the operator in selecting the most acceptable color match with the tooth/teeth in question. Their use depends on the adequacy of the shade guide and the familiarity of the operator with the color matching procedures. The requirements for any shade guide include: (1) A logical arrangement of colors in the color space, and (2) an adequate distribution of colors in color space. An ideal color space is one in which each color is the center of a sphere, and the next closest one matches the color surrounding it.

Most manufacturers provide shade guides for their material and it is advisable that these should not be used for shade matching of materials from other manufacturers. Many shade guide systems use hue as the basis to define color. Within each hue, there are various levels of chroma (or saturation). The third and most critical component for color selection (value) has no accountability and is not addressed in most shade guides. The VITA Classical Shade Guide is the most commonly used shade guide; however, it does not address value. Tabs are grouped into letters as:

A (red-yellow hue) : A1 A2 ….. so on
B (yellow hue) : B1 B2 ….. so on
C (grey hue) : C1 C2 …… so on
D (red-yellow grey) : D1 D2 ……. so on
Chroma is designated with numbers 1, 2, 3, 4.

The VITA 3D Master Shade Guide revolutionized shade selection, and made the value as the base of color. With the 3D-MasterGuide, as each color component is selected, it is held constant while the next color component is found. The traditional VITA Classical Shade Guide has 16 shade tabs and can be difficult for proper shade selection. The VITA 3D-Master allows for considerably more interpretation. VITA has also introduced the VITA Linear-guide, which simplifies the arrangement of the original VITA 3D-Master by separating the value tabs into their own holders. The chroma and hue tabs are arranged separately within each value group. This allows quick determination of the value before moving onto chroma and hue (Fig. 24.20). The tabs are grouped into five categories (number in sequence) with increasing value (1, 2, 3, 4, 5). Hue is designated with letters L, M and R. Chroma is also designated as 1, 2, 3, 4, 5 (For example, 3M2 correspond to value 3 and hue M and chroma 2). Chromascope uses different numbers to distinguish hue [100 (white), 200 (yellow), 300 (orange), 400 (green), 500 (brown)]. Chroma is also designated in numbers (10, 20 … so on).

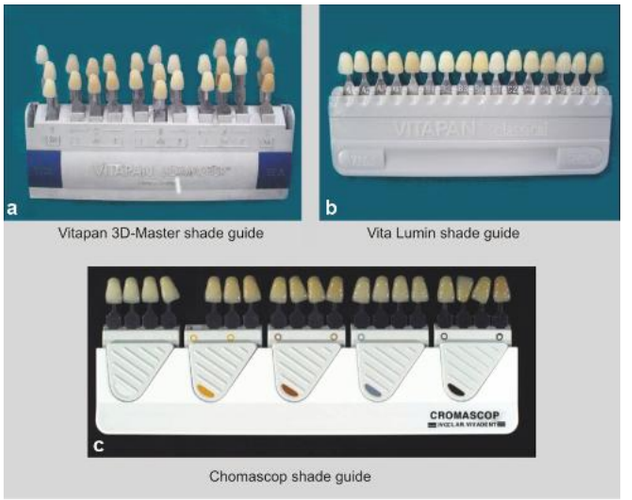


Fig. 24.20: Shade guides

Shofu offered the Natural Color Concept (NCC). Egger conducted a worldwide clinical study in which the tooth shades of more than 3500 individuals were recorded and analyzed spectrophotometrically. The NCC color indicator, derived from these data, served as basis for the dental color space. The NCC system consists of 208 color blends based on 38 basic shades. These blends are arranged in L*a*b* color space.

Shade guides are of two types: Key ring guides and metal slotted type fixed in plastic holders (mostly preferred).

Limitations of Shade Guide

- Because of reflectance, refractiveness, opacity, translucency and transparency, a given shade button may not match the tooth color.
- Thickness of the porcelain tab may be different from that of the prosthesis to be made (show different color).
- Porcelain shade from the same manufacturer may vary from batch to batch.
- Variation in surface texture of a tooth and shade button may affect color matching.
- Most shade buttons have a shape of maxillary central incisor, introducing another variable in shade matching.
- The necks of the shade tabs are usually darker in shade; need to be ground off for proper matching.
- The 'numbered' information sent to laboratory may vary in color of their shade guide.
- Light is reflected and transmitted through shade guide tabs (look translucent); whereas in restoration, light is reflected and not transmitted (look opaque and dense).



Fig. 24.21: Shade-taking devices: **(A)** Vita easyshade; **(B)** Shade eye NCC; **(C)** Shadescan; **(D)** Shaderie dental vision system; **(E)** Spactroshade

SHADE-TAKING DEVICES

Various devices have been designed to aid clinicians and technicians to specify and manage tooth color (Fig. 24.21).

The common devices are as follows.

a. Colorimeters

Colorimeters generally use three or four silicon filters that simulate the standard observer functions. The filter colorimeters are considered inferior to scanning devices, because of their inability to match the observer function, especially at low light levels. However, the device is used for quality control.

b. Digital Cameras (Filter Colorimeter)

The digital camera captures images using CCDs, which contain millions of microscopically small light-sensitive elements (photosites). There are several ways of recording the three colors in a digital camera; mostly use three separate sensors, each with a different filter. Light is directed to the different filter/sensor combinations by placing a beam splitter in the camera. The beam splitter allows each detector to see the image simultaneously.

c. Scanning (Spectrophotometers and Spectroradiometers)

There are two types of designs commonly used for scanning. The traditional scanning instrument consists of a single photodiode detector that records the amount of light at each wavelength. The light is divided into small wavelength intervals by passing through a monochromator. The present design uses a diode array with a dedicated element for each wavelength. This design allows for the simultaneous integration of all wavelengths. Both designs are slower than digital cameras; but remain the preferred tool to develop accurate color-measuring devices.

Advantages of digital shade matching

- Shade matching is objective; can be verified repeatedly.
- Surrounding environment do not influence shade matching.
- Provide exact information; no subjectivity
- Color replication in the laboratory is accurate.
- Less chairside time; less fatigue.

Limitations of digital shade matching

- Accuracy may get affected by phenomenon of edge loss
- Position of mouthpiece can change color measurement, when repeated
- Laboratory should have the same system
- Expensive

The difference between colorimeter and spectrophotometer is tabulated in Table 24.7.

GUIDELINES FOR CLINICAL SHADE SELECTION

The features involved in clinical shade selection are as follows:

1. Lighting of the Operatory

The optimal light source in the northern sunlight in the middle of the day with a slight overcast sky (intensity—1500). This light contains almost equal blend of all wavelengths of light (Morning and evening exposures are rich in blue wavelength and reddish yellow wavelength, respectively).

Since, the intensity of daylight varies, the systems having the same spectrum as natural daylight have been proposed. The capacity of an artificial light to reproduce natural light is measured by using color-temperature and color rendering index (CRI).

White light, which is considered the best source for color comparison, is assigned a color rendering index (CRI) of 100. Any artificial sources with an index over 90 are considered adequate.

The color corrected fluorescent lamps, with a CRI of over 90, are available that provide optimal color matching. Chair light is not recommended for color-matching procedures.

Before proceeding for color selection, the patient's profession should be ascertained, e.g. whether the person works indoors under fluorescent lamps? Whether the patient is an actor and works on stage? Whether he/she is exposed more to sunlight? Whether he/she is a night club performer? The lights should be selected, which closely simulate the working conditions of the patient, e.g. a night club performer should be viewed under incandescent and fluorescent light sources.

Table 24.7: Difference between colorimeter and spectrophotometer

Colorimeter	*Spectrophotometer*
• Provides measurements that correlate with human eye–brain perception.	• Provides wavelength by wavelength spectral analysis of the reflecting and/or transmitting properties of objects without human interpretation
• Consists of sensor and simple data processor	• Consists of sensor plus data processor or computer with software
• Has a set illuminant and observer combination (usually 2°)	• Has many illuminant/observer combinations
• Isolates a broad band of wavelength (use tristimulus absorption filter)	• Isolates a narrow band of wavelength (use a prism or interference filter)
• Easy instrument	• More complex instrument
• Works well for routine comparisons of similar colors	• Works well for color formulation, measurement of metamerism and variable illuminant/observer conditions.
• Optimal for quality inspection	• Optimal for quality inspection and research and development as well.

Intensity of light is also important. The ratio of intraoral to extraoral illumination should be 3 to 5:1 with extra oral light intensity of 200–300 foot candles. This amount of lighting also reduces eye fatigue by decreasing the difference in brightness levels of the light needed for execution of intraoral procedures.

The location of light can be from ceiling or surface mounted installation.

The luminous ceiling is achieved by illuminating the ceiling by fluorescent lighting.

Open perimeter surface mounted installation involves using fluorescent fixtures only at the perimeter of the ceiling and not in the center. Open perimeter recessed installation restricts lateral and upward loss of light by providing a proper reflective surface to project the light downwards.

2. Surrounding Environment

Bright-colored surroundings influence the reflected light and interfere with color matching. The bright-colored curtains, wall paints, etc. should be avoided in the dental operatory. A drape can be used to mask an undesirable color in the patient's clothing, provided the drape itself does not introduce an undesirable reflecting color. Lipstick should be removed so that it does not affect the perceived color. Light gray is an ideal background for color matching. Surfaces with high gloss produce disturbing glares and should be avoided.

3. Tooth Condition

The concerned tooth and the adjoining teeth on both sides should be free of surface stains prior to color matching. The tooth should be moist while selecting the shade as dehydration results in a whiter appearance. Dehydration also results in loss of water from enamel, increasing the internal scattering of light and consequently the amount of reflected light reaching the operator. Dehydration becomes a specific problem when considerably long time is spent in arriving at a final color match. Under such situations, the patient is suggested to close the mouth between comparisons so that the teeth are moistened by running the tongue over them.

Shade matching should be carried out before the application of rubber dam. It prevents any light reflected from the dam to interfere with shade selection and provides a more natural environment. The isolated tooth may take at least 30 minutes to regain its original color.

Shade matching: Clinical guidelines

- The tooth surface is cleaned.
- Lipstick is to be removed.
- Bright-colored clothes should be covered with neutral colored drapes.
- Surface texture, gloss, opacity, etc. should be noted.
- Shade selection should be carried out in the beginning, before any eye fatigue (do not see for more than 5 seconds).
- A distance of 12–16 inches is recommended for shade matching (More distance reduces visual angle, thus decreasing the visual perception).
- Value is analyzed first, followed by hue and chroma.
- Shade should be finalized after one or two appointments by two observers.
- Do not match the shade of dry tooth or tooth immediately after bleaching
- Use gray color to rest the eyes, if fatigued (avoid blue colour in between shade matching to rest the eyes).

4. Patient and Operator Position

The patient should be seated upright with the oral cavity at the observer's eye level. The operator should stand directly in front of the patient when selecting a shade.

The shade selection is carried out at 12–16 inches distance (Distance from which the patient's teeth are usually observed).

5. Time of Selection

The color selection process should be performed when the operator is able to devote sufficient time. The shade matching should be performed at the beginning of the day when eye fatigue has not set in.

PROCEDURE FOR SHADE MATCHING IN PORCELAIN RESTORATIONS

The shade guide is arranged according to four basic hues: A, B, C and D (Table 24.8). In each hue, there are divisions like A1, A2, A3, A3.5 and A4, which represent different chroma, and similar divisions in B, C and D. Another less common arrangement in shade guide is according to the value: B1, A1, B2, D2, C1, C2, D4, A3, D3, A3.5, B4, C3, A4 and C4.

The first step is to select the 'hue'. Since there is not much difference amongst the different hues, and to simplify the procedure, 'the four hues technique' is used. In this, the maximum chromas of each hue, i.e. A4, B4, C4 and D4, are selected and placed at equal distances. This permits effectively visualize the differences in hue because chromas of maximum intensity are placed besides each other. The shade is selected in no more than five seconds to avoid fatigue of cones in the retina.

After the basic hue is determined, a color that closely matches the tooth, i.e. chroma is chosen. For this, all the different chromas of the selected tab, say B, i.e. B1, B2, B3, B3.5 and B4, are placed one by one alongside the tooth to be restored. For simulating natural conditions more closely, the shade tab is covered with the patient's lips or operator's thumb to create a shadow effect. A general guideline that may assist the clinician while selecting a chroma is:

- Maxillary incisors and premolars are similar in chroma, with the central incisors being the lightest.
- Mandibular incisors are generally one chroma lighter than the maxillary incisors.
- Maxillary canines are at least two chroma darker than the maxillary incisors (applies particularly to the dentin shade).

The second shade guide (arranged according to value) is used to select the value. Avoid considering the hue and chroma when choosing for a value. Squinting aids in selection as it sensitizes the more sensitive black and white rods. The value is selected and the results again evaluated by matching against the corresponding opaque shade.

Table 24.8: Four basic hues in the VITA shade guide

Groups	*Hue*
A	Brownish tones
B	Yellowish tones
C	Gray tones
D	Reddish tones

Surface Texture

A smooth surface reflects most of the incident light back to the observer; whereas, a roughened surface scatters light in many directions.

Newly erupted teeth have a rough surface texture.

As the age advances, the surface features are worn down and the tooth appears smooth with a highly reflective glossy surface. The surface characteristics should be simulated for a particular age group so as to achieve a life-like restoration.

Esthetics Prescription Form

The esthetics prescription form contains detailed directions about the color. An elaborate diagram (Fig. 24.22) describes the location of the different characteristics, like incisal translucency, areas of high chroma, hypocalcified areas, craze lines and stains. If additional depth of translucency is required, it is marked as 'overlay' of transparent porcelain. An increase in the thickness of enamel porcelain may alter the shade considerably, whereas a thin layer of overlying porcelain will increase the depth of translucency with little alteration in the shade. The translucency of the incisal tip can be accentuated by placing translucent porcelain behind the incisal edge. 'Stain colors' can be marked to represent areas of craze lines, hypocalcified areas, etc.

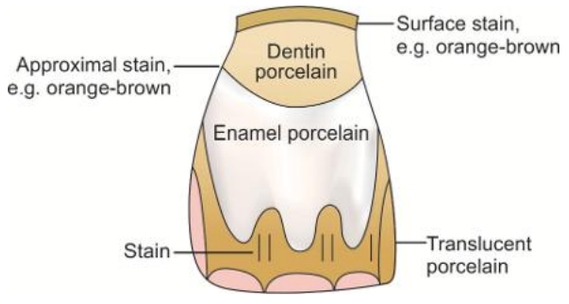


Fig. 24.22: Location and pattern of different characteristics, like, incisal translucency, craze lines and stains (diagrammatic)

Identification Tab

A tab mold using maxillary incisor is fabricated, coating a thin layer of porcelain on the mold. The selected opaque, dentin and incisal porcelains are then layered in prescribed thickness. Characterizations, such as, deep translucency, mamelons, craze lines and hypocalcification can be incorporated into the identification tab.

Characterization Map

An exact location of hypocalcification/translucency pattern is recorded for duplication in a ceramic restoration. The hypocalcification or translucency patterns are drawn on the contralateral tooth of the working side using Boley gauge. An alternative procedure is to paste transparent tape on the contralateral tooth and draw the desired pattern on the tape.

Staining/Modification of Stain

Ceramic stains used for surface modifications are usually metallic oxide pigments. These are similar in composition to the colorants used in opaque and body porcelains.

Stains can be applied on the glazed or unglazed porcelain. When applied to an unglazed surface, the stain penetrates to a much greater depth creating a more diffused appearance. For subdued changes, the stain may be applied directly over the glaze. No matter to what surface the stain is applied, it should be absolutely clean and devoid of any contaminants.

The modification of shade is carried out by the application of different stains. The stain powder is mixed with enough liquid medium to produce a heavy creamy consistency. A small increment of stain is picked up with the brush tip and applied over the surface. Any excess is removed with the brush.

The foremost rule for shade modification is 'moderation'. Only minor changes should be attempted. The operator should avoid changing the basic hue as he/she may lose control over the selected color. Complementary hue reduces both the chroma and value with a minimal loss in translucency. An over addition of complement also change the basic hue.

BIBLIOGRAPHY

1. Bayindir F, Gozalo-Diaz D, Kim-Pusateri S, Wee AG. Incisal translucency of vital natural unrestored teeth:A clinical study. J Esthet Restor Dent 2012; 24:335–39.
2. Bona AD, Barrett AA, Rosa V, Pinzetta C. Visual and instrumental agreement in dental shade selection: Three distinct observer populations and shade matching protocols. Dental Materials 2009;25:276–80.
3. Brewer JD, Wee AS. Advances in color matching. Dent. Clin. North Am 2004;48:341–45.
4. Burkinshaw SM. Color in relation to dentistry. Fundamentals of color science. Br Dent J 2004;33:196–200.
5. Cal E, Guneri P, Kose T. Comparison of digital and spectrophotometric measurements of color shade guides. J Oral Rehabil 2006;33:221–25.
6. Cal E, Sonugelen M, Guneri P, Kesercioglu A, Kose T. Application of a digital technique in evaluating the reliability of shade guides. J Oral Rehabil 2004;31:483–87.
7. Dagg H, O'Connell B, Claffey N, Byrne D, Gorman C. The influence of some different factors on the accuracy of shade selection. J Oral Rehabil 2004;31:900–04.
8. Denissen H, Dozie A. Photometric assessment of tooth color using commonly available software. Eur J Esthet Dent 2010;5:204–07.
9. Fondriest J. Shade matching in restorative dentistry: The science and strategies. Int J Periodont Restor Dent 2003; 23:467–71.
10. Hindle JP, Harrison A. Tooth color analysis by a new opto-electronic system. Eur J Prosthet Rest Dent 2000; 8:57–60.
11. Igiel, Lehmann KM, Ghinea R, Weyhrauch M, Hangx Y, Scheller H, Paravina RD. Reliability of visual and instrumental color matching. J Esthet Restor Dent 2017; 29:303–08.
12. Joiner A. Tooth color: a review of the literature. J Dent 2004;32:3–12.
13. Lehmann KM, Devigus A, Igiel C, Weyhrauch M, Schmidtmann I, Wentaschek S, Scheller H. Are dental color measuring devices CIE compliant? Eur J Esthet Dent 2012;7:324–33.
14. Mayekar S. Shades of a color: illusion or reality? Dent. Cl. N Am. 2001;45:155–59.
15. Moser JB, Wozniak TP, Moore BK. Use of Munsell system to complete color differences in composite resins. J Dent Res 1978;57:958–62.
16. Odiara C, Itoh S, Ishibashi K. Clinical evaluation of a dental color analysis system: The Crystaleye Spectro-photometer. J Prosthet Research 2011;55:199–201.
17. Pecho OE, Perez MM, Ghinea R, Bona AD. Lightness, chroma and hue differences on visual shade matching. Dent Mater 2016;32:1362–73.
18. Ragain JC. A review of color science in dentistry: colorimetry and color space. J Dent Oral Disord Ther 2016;4:1–5.
19. Ragain JC. A review of color science in dentistry: the process of color vision. J Dent Oral Disord Ther 2015;3:1–4.
20. Reno EA, Sunberg RJ, Black RP, Bush RD. The influence of lip/gum color on subject perception of tooth color. JDR 2000;79:381–84.
21. Russell MD, Gulfraz M, Moss BW. In vivo measurement of color changes in natural teeth. J Oral Rehab 2000; 27:786–90.
22. Shammas M, Alla RK. Color and shade matching in dentistry. Trends Biomater. Artif Organs 2011;25:172–75.
23. Sikri V. Color: Implications in dentistry J Conserv Dent 2010;13:249–55.
24. Smith AJ, Savitha PN. Shade matching in aesthetic dentistry-from past to recent advances. J Dent Oral Care Med 2017;3:1–9.
25. Sproull RC. Color matching in dentistry-Part I. The three dimensional nature of color. JPD 2001;86:453.

26. Stephen J. Precision shade technology; contemporary strategies in shade selection. Prac Proced Aesthet Dent 2002;14:79–83.
27. Takatsui F, Ferrarezi de Andrade M, Neisser MP, Barros LAB, de Castro Monteiro Loffredo L. CIE L*a*b*: comparison of digital images obtained photographically by manual and automatic modes. Braz Oral Res 2012;26:578–83.
28. Villavicencio-Espinoza CA, Narimatsu MH, Furuse AY. Using cross-polarized photography as a guide for selecting resin composite shade. Oper Dent 2018;43:113–20.
29. Yuan JCC, Brewer JD, Monaco ED, Davis EL. Defining a natural tooth color space based on a 3-dimensional shade system. J Prosthet Dent 2007;98:110–13.